Democracy and Economic Growth: Transitions to Democracy

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Abstract

This paper proposes an overlapping-generations model in which the redistribution rate of the economy affects both physical capital accumulation, through a tax on bequests, and human capital accumulation, through public education financed by that tax. The production function has constant elasticity of substitution between physical and human capital. We show that preferences for redistribution depend only on whether the two inputs are gross substitutes or gross complements: with an elasticity of substitution above one, workers choose a higher rate than the elite; with Cobb-Douglas technology, both groups choose the same rate; and with gross complements the ordering reverses. We then analyze an exogenous democratic transition, understood as a shift from the elite's to the median worker's preferred rate. Under gross substitutes, the transition lowers physical capital and raises human capital, in line with the within-country evidence documented for a panel of 93 countries between 1970 and 2005. The net effect on long-run income is negative in every parameterization considered: the elite, as a residual claimant on capital accumulation, chooses a rate close to the one that maximizes steady-state wealth, whereas workers do not fully internalize the dynamic cost of taxing bequests. The model thus rationalizes the two factor-accumulation channels emphasized by the empirical literature, but it also shows that redistribution towards public education alone is not sufficient to generate long-run income gains from democratization.

1 Introduction

Since the 1990s, the literature has increasingly emphasized the role of institutions in shaping long-run economic development. This is exemplified by the seminal contributions of North (1990) and Acemoglu et al. (2001). The present paper examines the effects of a particular institution – democracy – and the channels through which it affects income, particularly by shaping incentives to accumulate physical and human capital.

We propose an overlapping generations model with two types of agents: capitalists and workers. Capitalists' income originates from bequests received from their parents; they accumulate physical capital to be left for their offspring. Workers accumulate human capital through public education and earn labor income. A government taxes bequests (physical capital) to finance education spending. Output is generated by combining physical and human capital. Crucial to our analysis, we suppose a CES production function and focus on the gross substitutes case, that is, the elasticity of substitution is greater than one between the two inputs.

We first analyze an equilibrium with elite ruling, with the tax rate determined by capitalists. Then we study a democratic transition, in which fiscal policy is set by the median voter – which is a worker. The model predicts an increase in capital taxation and public spending on education. Numerical simulations indicate that the stock of physical capital tends to decrease, while human capital tends to rise. These effects are broadly in line with within-country correlations documented here for a panel of 93 countries between 1970 and 2005. Upon removing country fixed effects, democracy is positively correlated with human capital investment and negatively correlated with physical capital investment.

Our analysis shows that the gross substitutes assumption is crucial to reproduce these patterns. When technology is Cobb-Douglas, the two groups agree on the redistribution rate, and democracy does not affect factor accumulation; and when the inputs are gross complements, the ordering reverses: capitalists favor more redistribution than do workers.

The mechanisms emphasized here were already discussed by literature, although not in a unified framework such as ours. For instance, Przeworski and Limongi (1993) argues that democracy could discourage physical investment, as it generates greater social pressure for redistribution and taxation. This argument is also formalized by Persson and Tabellini (1994) and Alesina and Rodrik (1994). However, these authors do not assess the effect that such redistribution would have if reinvested in productive activities, rather than merely distributed in a lump-sum fashion.

The effect on human capital accumulation is emphasized by Saint-Paul and Verdier (1993). The argument holds that access to the political system allows individuals to demand more education, thereby increasing the economy's overall human capital stock, while under a non-democratic regime the demand for education is suppressed. They present a model in which individuals are heterogeneous in their human capital. Human

capital income is taxed and redistributed equally among all agents. The main conclusion is that democracy enhances redistribution, causing the economy to grow faster compared with authoritarian regimes. However, the authors do not integrate this channel with the effect on physical capital.

Here we combine the two effects discussed above and deliver implications for both physical and human capital. Our model is based on Galor and Moav (2006)[1], with an important difference: the production function has a constant elasticity of substitution (CES) between physical and human capital rather than a Cobb-Douglas form.[2] This difference turns out to be essential. With Cobb-Douglas technology, factor shares are constant and every group in the economy wants the redistribution rate that maximizes output; the elite and the workers then agree on the rate, and a political transition changes nothing. With a CES technology, the income share of each factor depends on the factor mix, so each group also cares about how redistribution tilts that mix. Only when physical and human capital are gross substitutes, do workers prefer a higher redistribution rate than the elite.

On the empirical side, the evidence points to a positive effect of democracy on human capital and a negative, or at most weak, effect on physical investment. Tavares and Wacziarg (2001) build a system of equations to identify the effect of democracy on the proximate determinants of growth, and find a positive effect on human capital and a negative effect on investment, with a small negative net effect on growth. Acemoglu et al. (2015) revisit the relationship between democracy, redistribution, and inequality using a broad panel of countries, and find a significant and robust effect of democracy on tax revenues as a fraction of GDP, consistent with the redistribution mechanism formalized in this paper, while documenting that the ultimate impact on inequality is more complex than simple median-voter models predict. Acemoglu et al. (2019) provide causal evidence that democratization increases GDP per capita by approximately 20 percent in the long run, using a dynamic panel strategy that controls for country fixed effects and leaverages on regional democratization waves as instruments; their results suggest that the income gains operate through greater investment in education, health, and physical capital, as well as through reduced social unrest. Rodrik and Wacziarg (2005), Giavazzi and Tabellini (2005), Persson and Tabellini (2008), and Papaioannou and Siourounis (2008) also find a positive effect of democracy on economic growth.

In Section 2, we revisit this evidence with a balanced panel of 93 countries observed every five years between 1970 and 2005. Once country fixed effects are removed, democracy is negatively associated with the investment share and positively associated with human capital, while the association with income is positive but considerably attenuated

[1]Other important contributions characterize how non-democratic regimes use barriers to entry in markets to protect incumbents, thereby discouraging technological progress. Among these, see Acemoglu (2008) and Aghion et al. (2008).

[2]We also add to them by studying the effect of democratic transitions

relative to the pooled cross-section, as in Acemoglu et al. (2008). An event study around 81 democratic transitions confirms the timing: human capital rises after transitions, the investment share responds weakly, and income rises moderately, although both income and human capital already display upward pre-trends.

Notheless, in the model, income tends to go *down* as a result of a democratic transition. This follows because the elite, who own the capital stock, choose the redistribution rate that maximizes the after-tax return on bequests, which is also the rate that maximizes steady-state wealth. Workers, who own no capital, do not fully internalize the dynamic cost of taxation and choose a rate on the other side of the output-maximizing rate. Redistribution towards public education therefore reproduces the factor-accumulation pattern in the data, but it cannot by itself account for the long-run income gains documented by Acemoglu et al. (2019); those must operate through channels absent from the model, such as productivity, reduced rent extraction, or lower social conflict.

The paper is organized as follows: Section 2 presents the stylized facts; Section 3 lays out the model; Section 4 analyzes the political regimes and the choice of redistribution rate; Section 5 presents the main simulation results; Section 6 carries out a robustness analysis; and Section 7 concludes.

2 Stylized Facts

This section presents the empirical evidence on the relationship between democracy and economic performance using data from the original sources employed in the literature. Recognizing that pooled cross-sectional regressions confound within-country dynamics with time-invariant country heterogeneity, a concern raised by Acemoglu et al. (2008), who show that the positive income–democracy correlation largely disappears once country fixed effects are included, the analysis focuses on variation *within* countries over time, through two complementary exercises: a within-country demeaned analysis, which removes country fixed effects, and an event study centered on democratic transition episodes.

2.1 Data sources and variable construction

Per capita income and investment. Data on per capita income (denoted by y, in 2005 international dollars, chained series) and investment as a share of gross domestic product (denoted k, expressed as a percentage) are taken from the Penn World Table 7.1 (PWT 7.1; Heston et al., 2012). Five-year averages are computed for all variables, yielding observations for the years 1970, 1975, 1980, 1985, 1990, 1995, 2000, and 2005. Only countries with complete data on all variables throughout the entire period are retained,

leaving a balanced panel of 93 countries and 744 country-period observations.[3]

Human capital. Human capital is constructed following Hall and Jones (1999), as is standard in the development accounting literature. Let E_{it} denote the average years of schooling for country i in period t. Human capital is then given by:

$$h_{it} = e^{\varphi(E_{it})}, \tag{1}$$

where the function $\varphi(\cdot)$ is piecewise linear, with slope coefficients corresponding to the returns to education at each schooling segment, taken from the cross-country Mincerian regressions of Psacharopoulos (1994):

- **0–4 years:** $\varphi'(E) = 0.134$, based on evidence from sub-Saharan Africa;
- **5–8 years:** $\varphi'(E) = 0.101$, the worldwide average return;
- **above 8 years:** $\varphi'(E) = 0.068$, based on evidence from the OECD.

Data on average years of schooling by country are taken from the updated Barro–Lee dataset (Barro and Lee, 2013). The derivative $\varphi'(E)$ can be associated with the Mincerian wage return to education (Mincer, 1974): if $\varphi(E) = 0$ for all E, the model reduces to the case with undifferentiated labor.

Democracy. The democracy indicator (*polity2*) is taken from the Polity5 Project (Marshall and Gurr, 2020). In its original form ($p_{2,it}$), the variable ranges from -10 (fully authoritarian) to $+10$ (fully democratic). Following Acemoglu et al. (2008), it is standardized to the unit interval:

$$\widetilde{p}_{2,it} = \frac{p_{2,it} + 10}{20}, \tag{2}$$

so that $\widetilde{p}_{2,it} \in [0, 1]$, with 0 corresponding to the most authoritarian regime and 1 to the most democratic. Periods of political interruption, interregnum, or transition (coded as -66, -77, and -88 in the original data) are set to missing and excluded from the sample.[4] Throughout the remainder of this section, "*polity2*" refers to the standardized measure $\widetilde{p}_{2,it}$.

Summary statistics. Table 1 reports descriptive statistics for the resulting balanced panel. The mean and standard deviation of *polity2* (0.56 and 0.37, respectively) are very close to those reported by Acemoglu et al. (2008) for a comparable sample (0.57 and 0.38), providing confidence that our sample captures the same cross-country variation as

[3] The sample of 93 countries is determined by the intersection of coverage across the three data sources described in this section.

[4] The Polity5 dataset updates and extends the original Polity IV series (Marshall and Jaggers, 2007). For the 1970–2005 period analyzed here, the two datasets are substantively equivalent.

the prior literature. The large dispersion in per capita income (a standard deviation of \$8,659, larger than the mean of \$7,697) reflects the wide cross-country heterogeneity that motivates the use of within-country identification strategies in the analyses that follow.

Table 1: Summary statistics

Variable	N	Mean	Std. Dev.	Min	Median	Max	Source
Per capita income, y (USD)	744	7,697.00	8,658.89	277.16	4,127.47	47,150.30	PWT 7.1
Investment share, k (% GDP)	744	22.53	10.44	1.59	21.22	64.59	PWT 7.1
Avg. years of schooling, E	744	5.86	2.95	0.29	5.76	12.86	Barro–Lee
Human capital, h	744	2.09	0.60	1.04	2.04	3.56	Barro–Lee
Democracy, *polity2*	744	0.56	0.37	0.00	0.59	1.00	Polity5

Source: own elaboration. Data: PWT 7.1 (Heston et al., 2012); Barro–Lee (Barro and Lee, 2013); Polity5 (Marshall and Gurr, 2020). Five-year averages. 93 countries, 1970–2005. $h = e^{\varphi(E)}$, Psacharopoulos (1994) piecewise returns.

2.2 Analysis I: Within-country variation

A naive pooled scatter plot of *polity2* against economic outcomes conflates two distinct sources of variation: persistent differences across countries (richer and better-educated countries tend to be democratic for reasons unrelated to the causal channels of interest) and changes within a country over time. Acemoglu et al. (2008) argue that the positive cross-sectional correlation between income and democracy largely disappears once country fixed effects are included, suggesting that the pooled relationship is driven primarily by time-invariant country heterogeneity.

To isolate within-country variation, the country mean is subtracted from every variable. For a generic variable x:

$$\tilde{x}_{it} = x_{it} - \bar{x}_i, \qquad \bar{x}_i = \frac{1}{T}\sum_{t=1}^{T} x_{it}, \tag{3}$$

where $T = 8$ is the number of periods. The demeaned variables $\tilde{x}_{it}$ contain only within-country, over-time variation, the same variation exploited by a panel fixed-effects estimator.

Figure 1 displays the within-country demeaned scatter for the three outcome variables. Points are colored by period to indicate whether the within-country relationship is driven by particular periods or is distributed uniformly across time.

Table 2 reports the pooled correlation matrix for reference, while Table 3 reports the within-country correlation matrix. Three findings are noteworthy.

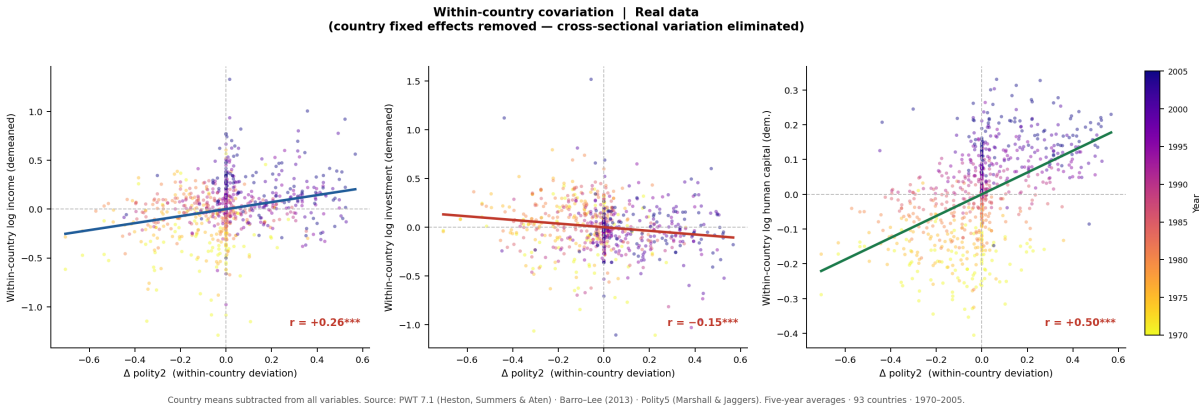


Figure 1: Within-country demeaned scatter: correlation between *polity2* and economic variables after removing country fixed effects. From left to right: demeaned log y, demeaned log k, demeaned log h. Each dot is one country-period observation; country means subtracted from all variables. Points colored by year. Fitted OLS line shown. Source: own elaboration. Data: PWT 7.1, Barro–Lee (2013), Polity5.

Table 2: Pooled correlation matrix

	y	h	k	*polity2*
y	1.000			
h	0.739***	1.000		
k	0.168***	0.180***	1.000	
polity2	0.563***	0.611***	0.005	1.000

$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$. Pearson correlations, pooled panel ($N = 744$).

Source: own elaboration. Data and sample as in Table 1.

Table 3: Within-country correlation matrix (country fixed effects removed)

	$\tilde{y}$	$\tilde{h}$	$\tilde{k}$	$\widetilde{polity2}$
$\tilde{y}$	1.000			
$\tilde{h}$	0.554***	1.000		
$\tilde{k}$	0.128***	−0.040	1.000	
$\widetilde{polity2}$	0.257***	0.496***	−0.149***	1.000

$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

All variables demeaned at the country level ($\tilde{x}_{it} = x_{it} - \bar{x}_i$). $N = 744$.

Source: own elaboration. Data and sample as in Table 1.

First, the within-country correlation between k and *polity2* is negative and highly significant ($r = -0.149$, $p < 0.001$), whereas the corresponding pooled correlation is

essentially zero ($r = 0.005$, $p = 0.892$). This sign reversal is a key empirical pattern motivating the model: once persistent cross-country differences are controlled for, democratization is associated with lower physical investment. This finding is consistent with Tavares and Wacziarg (2001), who use a structural-equations approach and find a negative effect of democracy on physical investment, and with the theoretical mechanism of Persson and Tabellini (1994) and Alesina and Rodrik (1994). It also aligns with the evidence in Acemoglu et al. (2015), who document that democracy significantly raises tax revenues as a share of GDP.

Second, the within-country correlation between h and *polity2* remains positive and statistically significant ($r = 0.496$, $p < 0.001$), confirming that the human capital channel operates even after removing country-level heterogeneity. This is consistent with Saint-Paul and Verdier (1993), who argue that democratic governments face greater pressure to invest in public education, and with the finding of Acemoglu et al. (2019) that the long-run income gains from democratization are associated with greater investment in education and health.

Third, the correlation between income and *polity2*, while still positive within countries ($r = 0.257$, $p < 0.001$), is considerably attenuated compared to the pooled estimate (0.563). This attenuation is consistent with Acemoglu et al. (2008), who show that much of the raw income–democracy correlation reflects common time-invariant factors rather than a causal effect of democratization on income.

2.3 Analysis II: Event study around democratic transitions

The demeaned analysis eliminates cross-sectional heterogeneity but cannot rule out omitted time-varying confounders, for instance a global growth wave that coincides with a wave of democratization. To examine the temporal ordering of the effects, an event-study design is employed, centering each country's time series on the period of a democratic transition and averaging across events. This approach has been applied to political-economy settings by Papaioannou and Siourounis (2008) and Persson and Tabellini (2008).

Identifying transitions. A democratic transition is defined as any five-year period in which *polity2* increases by more than 0.18 points, equivalent to a shift of more than 3.6 units on the original $[-10, 10]$ scale. This threshold is chosen to capture substantial, abrupt changes in political regime while excluding small fluctuations in the index. Applying this criterion to the balanced panel yields 81 transition events across the 1970–2005 period. The events are distributed across all decades in the sample, with a notable concentration in the period 1986–1995, reflecting the Third Wave of democratization described by Huntington (1991), which encompassed transitions in Latin America, Southern

Europe, Eastern Europe, and Sub-Saharan Africa.[5]

Normalized trajectories. For each transition event, the paths of $\log y$, $\log k$, and $\log h$ are extracted over a window of six five-year periods, from ten years before the transition ($t-10$) to fifteen years after it ($t+15$), where $t=0$ denotes the transition period. Each path is normalized to zero at $t=0$, so that $\Delta_s \equiv x_{t+s} - x_t$ measures the cumulative change relative to the transition period. The average of Δ_s across all 81 events is then plotted, together with a 95% confidence interval computed from the cross-event standard error.

Results. Figure 2 displays the event-study paths. Two pre-transition periods ($t-10$ and $t-5$) are shown to allow an assessment of pre-existing trends.

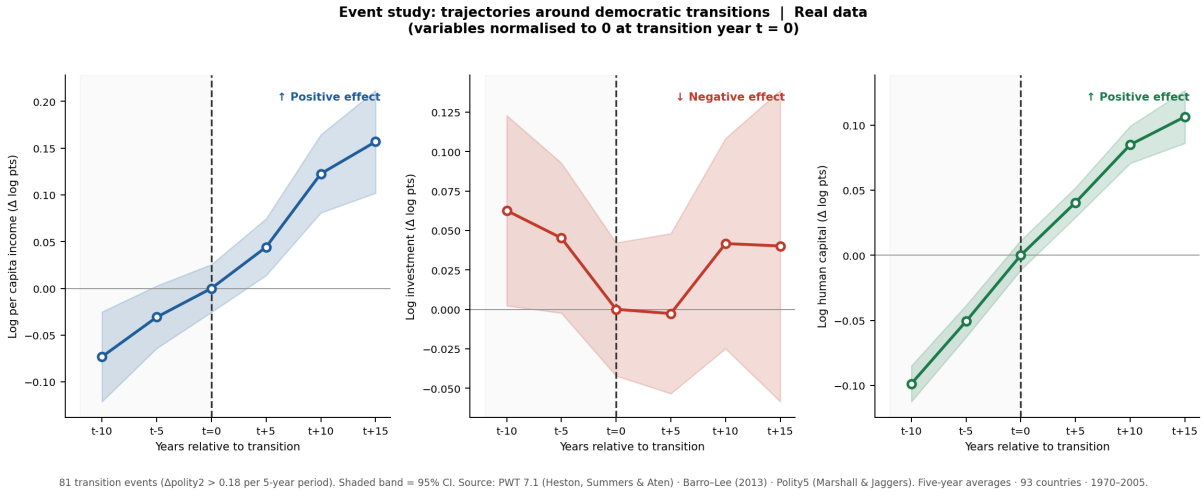


Figure 2: Event study: average trajectory of economic variables around democratic transitions ($t=0$). From left to right: log per capita income, log investment share, log human capital. Variables normalized to zero at $t=0$; horizontal axis in years relative to the transition period. Shaded band: 95% confidence interval around the cross-event mean. Grey shading: pre-transition window. $N=81$ transition events (Δ*polity2* >0.18 in one five-year period). Source: own elaboration. Data: PWT 7.1, Barro–Lee (2013), Polity5.

Three results stand out. First, the response of the investment share is weak. It is slightly negative in the first five years after the transition, but the decline is small relative to the confidence band and is not statistically distinguishable from zero; by $t+10$ the investment share has returned to a level comparable to the pre-transition one. The investment share was also declining before the transition, so the event study does not identify a clear break in investment at the transition date. The negative within-country association

[5] A full list of the 81 identified events, together with the corresponding changes in *polity2*, is available in upon request.

between democracy and investment documented in Section 2.2 therefore appears to be a low-frequency phenomenon rather than a sharp response to the transition itself.

Second, human capital rises immediately after the transition: the change at $t + 5$ is already positive with a confidence band that excludes zero, and the path continues to rise through $t + 15$. However, human capital was also rising before the transition at a comparable pace, so the event study does not detect an acceleration of human capital accumulation at the transition date. This is to be expected for a slow-moving stock such as average years of schooling of the adult population: higher enrollment induced by the new regime takes decades to be fully reflected in the stock, and the secular expansion of schooling in the sample period affects democratizing and non-democratizing countries alike.

Third, per capita income displays a positive trend after the transition, with the cumulative gain reaching about 0.15 log points by $t + 15$. The initial response is close to zero, and the gains materialize gradually, a pattern consistent with Acemoglu et al. (2019), who estimate that democratization raises GDP per capita by around 20 percent over a 25-year horizon with most of the gain materializing only after a decade. Income was also rising before the transition, although the post-transition slope is somewhat steeper.

Regarding pre-trends, then, only investment is approximately flat in the pre-transition window, while income and human capital display upward pre-trends. The event study should therefore be read as evidence on the timing and relative magnitude of the responses rather than as a causal estimate: a key result from the within-country analysis remains, namely that transitions are associated with income growth accompanied by the accumulation of human capital rather than of physical capital.

2.4 Discussion

Taken together, the two analyses paint a coherent empirical picture. The *within-country* analysis (Section 2.2) shows that, once the dominant cross-sectional variation is removed, democracy is negatively associated with physical investment ($r = -0.149$) and positively associated with human capital ($r = 0.496$). The sign for investment is the opposite of what a naive cross-country comparison would suggest, and confirms the relevance of the within-country channel emphasized by Tavares and Wacziarg (2001). The *event study* (Section 2.3) provides complementary evidence on timing: following democratic transitions, human capital keeps rising, income rises gradually, and the investment share responds weakly, with a small and statistically insignificant decline in the first five years.

These findings motivate the theoretical model developed in Section 3. The model delivers the two factor-accumulation regularities – a negative effect of democratization on physical capital and a positive effect on human capital – and makes explicit the condition on technology under which they arise. As will become clear, this condition is that phys-

ical and human capital be gross substitutes. The model also allows us to ask whether redistribution towards public education is, by itself, sufficient to generate long-run income gains; the answer turns out to be negative.

3 Model

The model developed in this section is based on Galor and Moav (2006), with the difference that the production function used here features a constant elasticity of substitution between physical and human capital, whereas they use a Cobb-Douglas function.

Consider an overlapping-generations closed economy composed of a population of unit mass with two types of agents: capitalists (the elite) and workers. The proportion of capitalists is $\varepsilon_e \in (0, 0.5)$ and that of workers is $1 - \varepsilon_e$, so that workers are always the majority. Agents within each group are identical, so we drop individual indices and refer to a representative capitalist and a representative worker. Both agents live for two periods. Capitalists are born with a bequest from their parents, which is taxed and invested; in the second period of life, they consume their capital income and decide how much to bequeath to their offspring. Workers use the first period to study (with the aid of government-financed public education), while in the second period they consume their labor income. The government taxes bequests, thereby affecting the economy's capital stock, and uses the proceeds to finance public education for workers. The redistribution rate τ_t is chosen by the political process described in Section 4.

Timing of taxation. A capitalist born in period t receives a bequest b_t from his parent. The bequest is taxed at the rate τ_t when transferred, in period t; the after-tax amount $(1-\tau_t)b_t$ is invested and becomes part of the physical capital stock used in the production in period $t+1$, earning the gross return r_{t+1}. The tax revenue collected in period t finances public education in period t for the workers born in that period, whose human capital becomes productive in period $t+1$. Thus, τ_t is a tax on wealth transfers, not on capital income, and the gross return r_{t+1} does not enter the government's revenue.

3.1 Capitalists

Following Galor and Moav (2006), in the first period of life, a capitalist born at t consumes a fraction of his parent's consumption.[6] In period $t+1$, the individual receives the gross return on the after-tax bequest, and decides how much to consume and how much to leave to his offspring. Regarding this decision, the capitalist's utility function is:

$$U\left(c^e_{t+1},\, b_{t+1}\right) = (1-\beta)\ln c^e_{t+1} + \beta \ln b_{t+1} \tag{4}$$

[6] This consumption does not affect any decision.

where c^e_{t+1} denotes consumption in period $t+1$, b_{t+1} denotes the bequest left to the child, and $\beta \in (0,1)$ is the relative weight assigned to the child's bequest in the utility function.[7] The budget constraint of a capitalist born in period t is:

$$c^e_{t+1} + b_{t+1} \leq r_{t+1}(1-\tau_t)b_t \tag{5}$$

where the left-hand side represents total expenditure, divided between consumption and the bequest left to his child, and the right-hand side is the gross return on the after-tax bequest received from his parent (r_{t+1} is the return per unit of capital; full depreciation at the end of the period is assumed). Figure 3 illustrates the sequence of events for a capitalist born at time t.

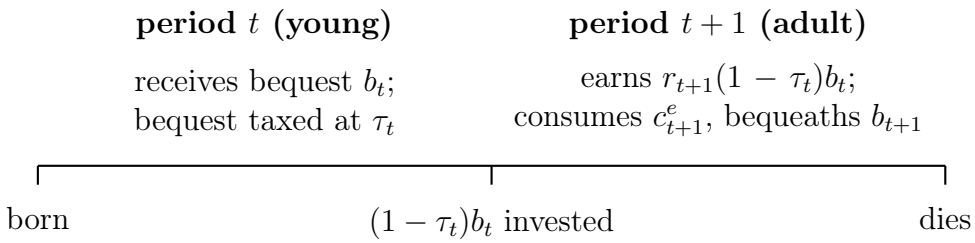


Figure 3: Capitalist's timeline

The capitalist's problem is therefore:

$$\max_{\{c^e_{t+1},\, b_{t+1}\}} (1-\beta)\ln c^e_{t+1} + \beta \ln b_{t+1} \quad \text{s.t.} \quad c^e_{t+1} + b_{t+1} \leq (1-\tau_t) r_{t+1} b_t$$

where τ_t, r_{t+1} and b_t are given.

The capitalist's consumption is given by equation (5) holding with equality, while the bequest is determined by the first-order condition:

$$\frac{1-\beta}{(1-\tau_t)r_{t+1}b_t - b_{t+1}} = \frac{\beta}{b_{t+1}} \tag{6}$$

In equation (6), the left-hand side is the marginal utility cost of reducing consumption, while the right-hand side is the marginal utility benefit of a larger bequest. From (6) one obtains:

$$b_{t+1} = \beta(1-\tau_t)r_{t+1}b_t \tag{7}$$

With log utility, a constant fraction β of after-tax capital income is bequeathed.

3.2 Workers

Figure 4 illustrates the sequence of events for a worker born at time t.

[7] The superscript e indicates elite.

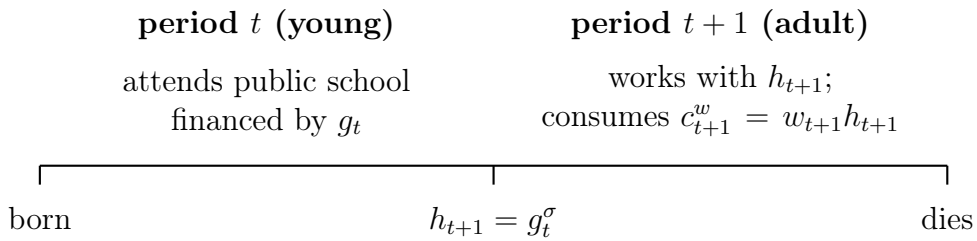


Figure 4: Worker's timeline

The first period of life (t) is dedicated to studying[8]. In the second period ($t+1$), the individual consumes his labor income. Therefore, the worker maximizes the utility function:[9]

$$U\left(c^w_{t+1}\right) = \ln c^w_{t+1} \tag{8}$$

The constraints of a worker born in period t are:

$$c^w_{t+1} \leq w_{t+1}\, h_{t+1} \tag{9}$$

$$h_{t+1} = h(g_t) = g_t^\sigma \tag{10}$$

where w_{t+1} is the wage per unit of human capital, h_{t+1} is the human capital accumulated in the first period, g_t is government spending on education per worker, and $\sigma \in (0,1)$.[10] Equation (9) is the worker's budget constraint, while equation (10) describes the human capital technology operated by the state.

3.3 Firms

Firms produce the final good using physical and human capital according to a CES production function:

$$Y_t = A\left[\alpha K_t^{-\rho} + (1-\alpha)H_t^{-\rho}\right]^{-1/\rho} \tag{11}$$

where $\rho \in [-1, 0)$, $\alpha \in (0,1)$, K_t is the total stock of physical capital, H_t is the total stock of human capital in the economy, and $A > 0$ is the economy's exogenous technology level. The elasticity of substitution between physical and human capital is

$$\varepsilon_{K,H} = \frac{1}{1+\rho},$$

so that with $\rho < 0$ the two inputs are gross substitutes ($\varepsilon_{K,H} > 1$). If $\rho > 0$, K and H are gross complements, and the limit $\rho \to 0$ corresponds to the Cobb-Douglas case $Y_t = AK_t^\alpha H_t^{1-\alpha}$. We maintain $\rho < 0$ as the baseline assumption; in Section 4 we

[8] Again, the individual consumes a fraction of his parent's income in the first period.

[9] The superscript w indicates worker.

[10] This implies $h'(g_t) > 0$ and $h''(g_t) < 0$.

show that this assumption is what determines the direction of the political conflict over redistribution, and in Section 6 we report simulations with $\rho \geq 0$ for comparison.

Since factor markets are competitive, the return to capital and the wage per unit of human capital are the marginal products:

$$r_t = A\alpha K_t^{-\rho-1}\left[\alpha K_t^{-\rho} + (1-\alpha)H_t^{-\rho}\right]^{-\frac{\rho+1}{\rho}} \tag{12}$$

$$w_t = A(1-\alpha)H_t^{-\rho-1}\left[\alpha K_t^{-\rho} + (1-\alpha)H_t^{-\rho}\right]^{-\frac{\rho+1}{\rho}} \tag{13}$$

Both r_t and w_t are homogeneous of degree zero in (K_t, H_t), and thus depend only on the ratio $\kappa_t \equiv K_t/H_t$. It will be useful to write the income share of physical capital as

$$s_K(\kappa_t) \equiv \frac{r_t K_t}{Y_t} = \frac{\alpha\kappa_t^{-\rho}}{\alpha\kappa_t^{-\rho} + (1-\alpha)}, \qquad s_H(\kappa_t) \equiv \frac{w_t H_t}{Y_t} = 1 - s_K(\kappa_t). \tag{14}$$

With $\rho < 0$, s_K is increasing in κ_t: when inputs are gross substitutes, the share of the factor that becomes relatively abundant rises. With $\rho > 0$ the opposite holds, and with $\rho = 0$ the shares are constant.

3.4 Government

The government does nothing other than collect the bequest tax and spend the entire revenue on public education. The government budget is balanced in every period:

$$\varepsilon_e \, \tau_t b_t = (1-\varepsilon_e) g_t \tag{15}$$

where the left-hand side is the total collected from capitalists in period t (there are ε_e capitalists, each transferring a bequest b_t) and the right-hand side is the total spent on universal public education for the $1-\varepsilon_e$ workers born in period t. Therefore, education spending per worker is:

$$g_t = \frac{\varepsilon_e}{1-\varepsilon_e}\,\tau_t b_t \tag{16}$$

3.5 Equilibrium

The physical capital stock available in period $t+1$ is the sum of the after-tax bequests invested in period t:

$$K_{t+1} = \varepsilon_e(1-\tau_t)b_t \tag{17}$$

Using equations (10) and (16) and aggregating over workers, the human capital stock in period $t+1$ is:

$$H_{t+1} = (1-\varepsilon_e)\, h_{t+1} = (1-\varepsilon_e)^{1-\sigma}\varepsilon_e^{\sigma}\,\tau_t^{\sigma} b_t^{\sigma} \tag{18}$$

Therefore, from equations (17) and (18), the factor ratio is:

$$\kappa_{t+1} \equiv \frac{K_{t+1}}{H_{t+1}} = \Gamma\,(1-\tau_t)\,\tau_t^{-\sigma}\,b_t^{1-\sigma}, \qquad \Gamma \equiv \varepsilon_e^{1-\sigma}(1-\varepsilon_e)^{\sigma-1}. \tag{19}$$

Since $\sigma < 1$, the factor ratio is increasing in b_t and decreasing in τ_t: a richer elite makes physical capital relatively abundant, and a higher redistribution rate makes it relatively scarce.

Finally, from equations (7) and (12), the bequest dynamics is:

$$b_{t+1} = \beta(1-\tau_t)\,r(\kappa_{t+1})\,b_t, \qquad r(\kappa) \equiv A\alpha\kappa^{-\rho-1}\left[\alpha\kappa^{-\rho} + (1-\alpha)\right]^{-\frac{\rho+1}{\rho}} \tag{20}$$

with κ_{t+1} given by (19). In the Cobb-Douglas limit $\rho \to 0$, equation (20) reduces to the power law $b_{t+1} = D(1-\tau_t)^{\alpha}\tau_t^{\sigma(1-\alpha)}b_t^{\alpha+\sigma(1-\alpha)}$ with $D \equiv \beta A\alpha\,\Gamma^{\alpha-1}$; for $\rho \neq 0$ the law of motion is not a power law, and the simulations in Section 5 use (20) directly.

Proposition 1 (Steady state)**.** *Fix $\tau \in (0,1)$ and let $\rho \in [-1, 0)$. If*

$$\beta(1-\tau)A\alpha^{-1/\rho} < 1, \tag{21}$$

then the bequest dynamics (20) *have a unique steady state $b^*(\tau) > 0$, and b_t converges monotonically to $b^*(\tau)$ from any $b_0 > 0$.*

Proof. See Appendix A.5. The idea is that the growth factor $\phi(b) \equiv \beta(1-\tau)r(\kappa(b))$ is strictly decreasing in b, diverges as $b \to 0$ and converges to $\beta(1-\tau)A\alpha^{-1/\rho}$ as $b \to \infty$, while $b\,\phi(b)$ is strictly increasing. □

Condition (21) reflects the fact that, if K and H are gross substitutes, the marginal product of physical capital is bounded below by $A\alpha^{-1/\rho}$ as the economy accumulates capital without bound. For the parameter values used below the bound is negligible ($\alpha^{-1/\rho} = 0.33^{20} \approx 2 \times 10^{-10}$ at $\rho = -0.05$) and the condition holds trivially; as $\rho \to 0$, the bound vanishes and the condition becomes vacuous.

4 Political Regimes

This section presents the political process through which individuals make decisions regarding the economy's redistribution rate, both under democracy and under the elite regime. First, the case of a capitalists' regime, or dictatorship, is presented. Then, the choice of redistribution rate under universal suffrage, i.e. democracy, is presented. Finally, the two rates are compared. In both cases, the redistribution rate τ_t is chosen at time t, when b_t is predetermined, by the group that holds political power, taking into account the effect of τ_t on next period's factor prices.

4.1 Elite Regime

In this case, the only voters are the capitalists. They choose the redistribution rate that maximizes their utility:

$$\max_{\tau_t}(1-\beta)\ln c_{t+1}^e + \beta \ln b_{t+1} \quad \text{s.t.} \quad c_{t+1}^e + b_{t+1} \leq (1-\tau_t) r_{t+1} b_t$$

Substituting the constraint and the first-order condition (7) into the utility function, the problem becomes:

$$\max_{\tau_t}\ (1-\beta)\ln[(1-\beta)(1-\tau_t)r_{t+1}b_t] + \beta\ln[\beta(1-\tau_t)r_{t+1}b_t] \ = \ \max_{\tau_t}\ \ln[(1-\tau_t)r_{t+1}] + \text{const.}$$

The capitalist simply maximizes the after-tax return on bequests. The first-order condition is:

$$-r_{t+1} + (1-\tau_t)\frac{\partial r_{t+1}}{\partial \tau_t} = 0 \tag{22}$$

The first term in equation (22) is the marginal cost to capitalists of raising the redistribution rate (a smaller after-tax bequest), while the second term is the marginal benefit (a higher return to capital, because public education raises the stock of human capital with which physical capital is combined).

Substituting equations (12), (17), and (18) into (22), one obtains:

$$\frac{\rho}{1-\tau_t} - W(\tau_t, b_t) = 0 \tag{23}$$

where

$$W(\tau_t, b_t) \equiv \frac{\rho+1}{\rho} \cdot \frac{\rho\alpha\varepsilon_e^{-\rho}(1-\tau_t)^{-\rho-1}b_t^{-\rho} - \rho\sigma(1-\alpha)(1-\varepsilon_e)^{-\rho(1-\sigma)}\varepsilon_e^{-\rho\sigma}\tau_t^{-\rho\sigma-1}b_t^{-\rho\sigma}}{\alpha\varepsilon_e^{-\rho}(1-\tau_t)^{-\rho}b_t^{-\rho} + (1-\alpha)(1-\varepsilon_e)^{-\rho(1-\sigma)}\varepsilon_e^{-\rho\sigma}\tau_t^{-\rho\sigma}b_t^{-\rho\sigma}} \tag{24}$$

Equation (23) is used to find the elite's optimal rate τ_t^e in the simulations. The derivation is in Appendix A.1.

4.2 Democracy

In this case, everyone votes on the redistribution rate. Since there are more workers than capitalists, the median voter is a worker. Therefore, the median voter's problem is:

$$\max_{\tau_t} \ln c_{t+1}^w \quad \text{s.t.} \quad c_{t+1}^w \leq w_{t+1}\, h_{t+1}$$

The first-order condition is:

$$\frac{\partial w_{t+1}}{\partial \tau_t}\, h_{t+1} + w_{t+1}\, \frac{\partial h_{t+1}}{\partial \tau_t} = 0 \tag{25}$$

The first term is the marginal cost, in terms of the wage per unit of human capital, of raising the redistribution rate (a smaller physical capital stock lowers the wage), and the second term is the marginal benefit of accumulating more human capital.

Substituting equations (13), (16), (17), and (18) into the first-order condition (25), one obtains:

$$-\frac{\rho\sigma}{\tau_t} - W(\tau_t, b_t) = 0 \tag{26}$$

with W as in (24). The derivation is in Appendix A.2.

4.3 Comparison of Redistribution Rates

Both groups solve a problem of the same form. Using the factor shares (14) and the fact that $r_{t+1}K_{t+1} = s_K Y_{t+1}$ and $w_{t+1}H_{t+1} = s_H Y_{t+1}$, the two objectives can be written as:

$$(1-\tau_t) r_{t+1} b_t = \frac{r_{t+1}K_{t+1}}{\varepsilon_e} = \frac{s_K(\kappa_{t+1})\, Y_{t+1}}{\varepsilon_e}, \tag{27}$$

$$w_{t+1} h_{t+1} = \frac{w_{t+1}H_{t+1}}{1-\varepsilon_e} = \frac{s_H(\kappa_{t+1})\, Y_{t+1}}{1-\varepsilon_e}. \tag{28}$$

That is, each group maximizes its own income, which equals the income share of the factor it owns times total output. This decomposition makes the political conflict transparent. Let τ_t^Y denote the redistribution rate that maximizes next period's output Y_{t+1}, given b_t.

Proposition 2 (Ordering of redistribution rates)**.** *For any $b_t > 0$, next period's output Y_{t+1} is single-peaked in τ_t, with maximizer $\tau_t^Y \in (0,1)$. Let τ_t^e and τ_t^w be interior solutions of* (23) *and* (26)*, respectively. Then:*

(i) if $\rho < 0$ (gross substitutes), $\tau_t^e < \tau_t^Y < \tau_t^w$;

(ii) if $\rho = 0$ (Cobb-Douglas), $\tau_t^e = \tau_t^Y = \tau_t^w = \sigma(1-\alpha)/[\alpha + \sigma(1-\alpha)]$;

(iii) if $\rho > 0$ (gross complements), $\tau_t^w < \tau_t^Y < \tau_t^e$.

Proof. See Appendix A.3. □

The intuition follows from (27) and (28). Both groups value a higher Y_{t+1}, but each also cares about its factor share. A higher τ_t lowers $\kappa_{t+1} = K_{t+1}/H_{t+1}$. Under gross substitutes, the share of the relatively abundant factor rises, so lowering κ_{t+1} transfers income share from capital to human capital: the elite wants less redistribution than the output-maximizing rate, and workers want more. Under Cobb-Douglas technology, the shares are constant and the conflict disappears: both groups want to maximize output and choose the same rate, so a political transition has no economic effect. Under gross complements, the share of the relatively *scarce* factor rises, and the ordering reverses: the elite would tax its own bequests more heavily than workers would, because making

physical capital scarce raises its share. The gross substitution assumption is therefore necessary for the model to deliver a higher redistribution following a democratic transition. In our simulations, we show that this tends to lower physical capital and increase human capital, consistent with the pattern found in the data.

Figure 5 illustrates the two first-order conditions (23) and (26) for $\rho < 0$. The green curve plots $W(\tau_t, b_t)$ for a given value of b_t, while the blue and red curves plot $-\rho\sigma/\tau_t$ and $\rho/(1-\tau_t)$, respectively. The intersection of the blue and green curves yields the worker's tax rate τ_t^w, while the intersection of the red and green curves yields the capitalist's tax rate τ_t^e. As $\rho \to 0$, the blue and red curves both collapse onto the horizontal axis and the two rates coincide at τ^{Cobb}.

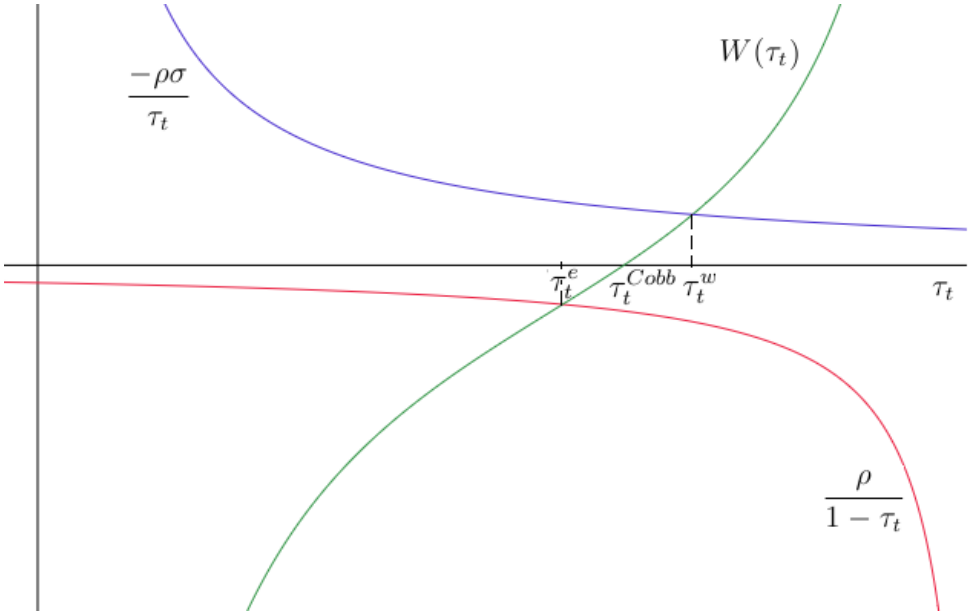


Figure 5: First-order conditions for the redistribution rate ($\rho < 0$)
Source: own elaboration.

Two further properties of the equilibrium will be useful for interpreting the simulations.

Proposition 3 (Neutrality of the elite's population share)**.** *The redistribution rates τ_t^e and τ_t^w, and the dynamics of $\tilde{b}_t \equiv \varepsilon_e b_t/(1-\varepsilon_e)$, do not depend on ε_e. Consequently, the steady-state values of K, H and Y are proportional to $(1-\varepsilon_e)$ under either regime, and the percentage changes in K, H and Y between the elite-regime and the democratic steady states are independent of ε_e.*

Proof. See Appendix A.4. □

The proportion of capitalists only scales the economy: a smaller elite with proportionally larger bequests generates the same factor ratio, the same factor prices, and the same political choices. This is why ε_e does not appear among the parameters varied in the robustness analysis of Section 6.

Proposition 4 (The elite's rate and steady-state wealth)**.** *Let $b^*(\tau)$ be the steady state of* (20) *under a constant rate τ, and let $\tau^e(b)$ denote the elite's rate when current wealth is b, assumed to be the unique interior solution of* (23)*. Then:*

(i) $db^*(\tau)/d\tau$ has the sign of $\tau^e(b^*(\tau)) - \tau$. In particular, $b^*(\tau)$ is a critical point of the function $\tau \mapsto b^*(\tau)$ at the elite's own steady-state rate τ^{e*}, defined by $\tau^{e*} = \tau^e(b^*(\tau^{e*}))$: $db^*/d\tau = 0$ there.

(ii) If $\rho < 0$, then at the democratic steady state ($\tau^{w*} = \tau^w(b^*(\tau^{w*}))$), wealth is strictly decreasing in τ: raising the rate above the workers' choice would lower long-run wealth even further.

(iii) If, in addition, the map $\tau \mapsto \tau^e(b^*(\tau))$ has slope strictly less than one everywhere, then $b^*(\tau)$ is single-peaked with its global maximum at $\tau = \tau^{e*}$: steady-state wealth, and hence steady-state physical capital, are lower under any constant rate above the elite's.

Proof. See Appendix A.6. □

The elite maximizes the after-tax return on bequests, which is precisely the growth factor of the bequest sequence. Since this growth factor is decreasing in b and equal to 1 in the steady state, the tax rate that maximizes the growth factor is also the rate that maximizes long-run wealth. The elite's rate $\tau^e(b)$ varies very little with b in all our numerical experiments (it falls by less than one percentage point per order of magnitude of b), so that condition (iii) holds and $b^*(\tau)$ is single-peaked at τ^{e*} in every simulation reported below. In other words, the elite chooses the tax rate that maximizes wealth.

Workers, on the other hand, take into account the effect of τ_t on next period's wage. However, since they do not own capital, they do not fully internalize the dynamic cost of taxation, specifically the effect of a lower after-tax return on the accumulation of bequests and, hence, on both factor stocks in all subsequent periods. As the simulations will show, this is the reason why steady-state income falls after a democratic transition even though human capital rises.

5 Simulation

This section analyzes the main quantitative implications of a democratic transition in the model. First, the values of the parameters are chosen. An initial bequest value b_0 is set, and it is assumed that the elite initially chooses the redistribution rate. The economy is then simulated until it reaches the elite-regime steady state. Next, an exogenous shock to the political regime (a democratic transition) is imposed at $t = 0$: from then on, workers choose the redistribution rate. Finally, the trajectories of the main variables are analyzed.

Table 4 reports the baseline parameter values. The capital share α and the bequest weight β are set to conventional values, and $\rho = -0.05$ implies an elasticity of substitution of about 1.05, within the range estimated by Duffy and Papageorgiou (2000). We stress that the parameter values are meant to illustrate the mechanisms of the model and the direction of the effects, as well as how they respond to changes in parameters; they are not a calibration designed to match the moments of Section 2, and the magnitudes reported below should not be interpreted as quantitative predictions. By Proposition 3, the values of ε_e and b_0 are immaterial for the percentage changes reported.

Table 4: Baseline Parameter Values

α	σ	β	ρ	A	ε_e	b_0
0.33	0.8	0.9	-0.05	10	0.2	1

Source: own elaboration.

Figure 6 shows the redistribution rate before and after the shock that occurs at $t = 0$. In the elite-regime steady state the rate is $\tau^e = 0.580$; after the transition, workers choose $\tau^w = 0.635$. Both are close to the Cobb-Douglas rate $\tau^{\text{Cobb}} = 0.619$ because ρ is small in absolute value, and they lie on opposite sides of it, as established in Proposition 2.

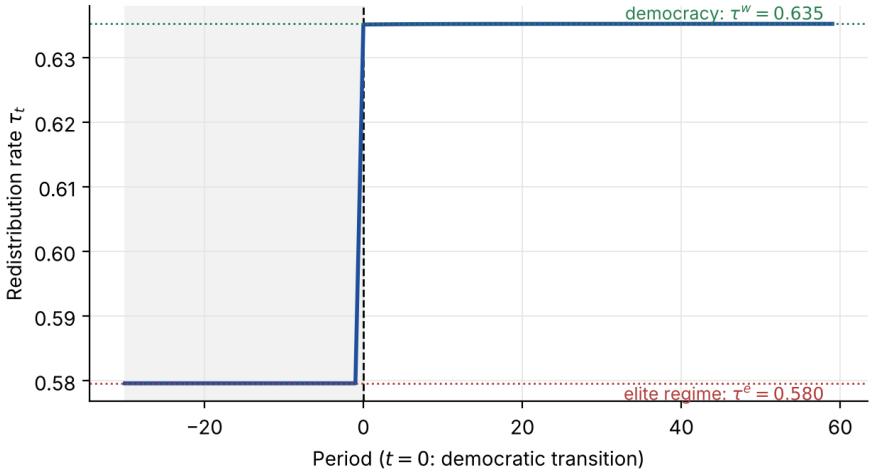


Figure 6: Redistribution rate before and after the transition

Source: own elaboration.

Figure 7 presents the trajectories of physical and human capital following the democratization shock. Elite-regime steady-state values are normalized to 100.

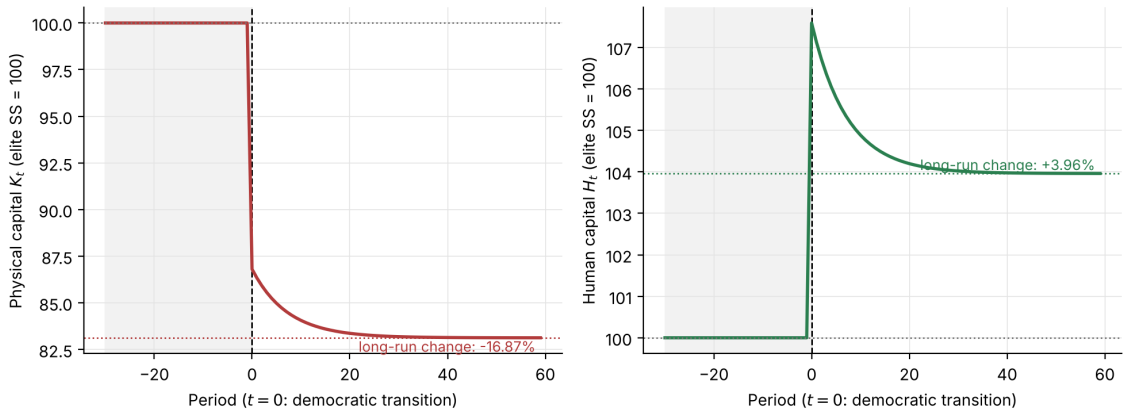


Figure 7: Physical and human capital trajectories
Source: own elaboration.

On impact, the higher redistribution rate lowers physical capital by 13.2 percent and raises human capital by 7.6 percent, since the after-tax bequest falls and education spending rises for a given b_0. Thereafter, bequests decline gradually towards the lower democratic steady state, because the after-tax return has fallen, and this erodes part of the initial gain in human capital while deepening the decline in physical capital. In the long run, physical capital is 16.9 percent lower and human capital 4.0 percent higher than in the elite regime. This result is in line with the stylized facts: under the democratic transition, physical capital falls and human capital rises.

Figure 8 shows the trajectory of income. Table 5 summarizes the percentage changes between steady states.

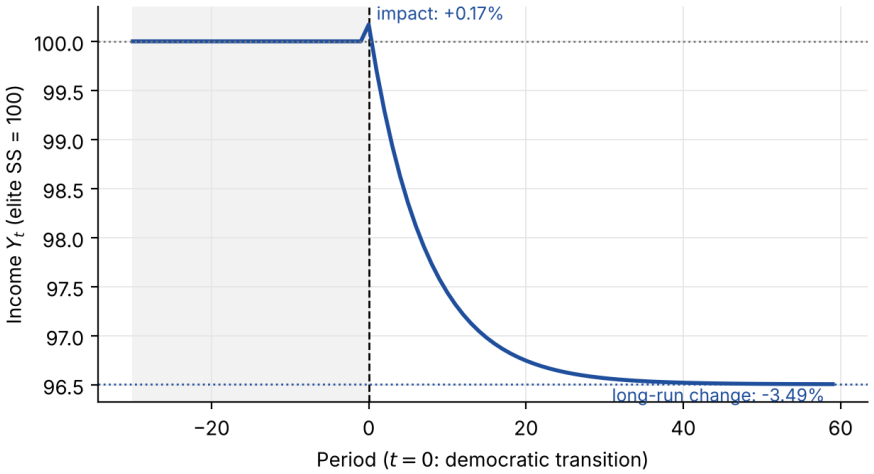


Figure 8: Income trajectory
Source: own elaboration.

Table 5: Percentage Change between Steady States (baseline)

Variable	On impact ($t = 0$)	Long run
Physical capital	-13.19	-16.87
Human capital	$+7.59$	$+3.96$
Income	$+0.17$	-3.49

Source: own elaboration. Changes in % relative to the elite-regime steady state.

Income is essentially unchanged on impact: the loss of physical capital and the gain in human capital almost exactly offset each other, because τ^e and τ^w lie on opposite sides of, and roughly equidistant from, the output-maximizing rate. Over time, however, income declines as bequests fall, converging to a steady-state level 3.5 percent below the elite regime's; half of the adjustment is completed within six periods. This is the content of Proposition 4: the elite's rate is (approximately) the rate that maximizes steady-state wealth, so any departure from it imposed by the median voter reduces long-run accumulation. In the baseline, the steady-state output-maximizing rate is 0.584, only slightly above $\tau^e = 0.580$ and well below $\tau^w = 0.635$.

The model therefore reproduces the two factor-accumulation facts of Section 2, but not the long-run income gains documented by Acemoglu et al. (2019). We return to this point in the conclusion.

6 Robustness Analysis

In this section, new simulations are performed varying the parameter values. Table 6 shows, in the first five columns, the parameter values used in each simulation; in the next three columns, the steady-state redistribution rates chosen by the elite and by the workers, and the income share of physical capital s_K in the elite-regime steady state; and, in the last three columns, the percentage change in physical capital, human capital, and income between the elite-regime and the democratic steady states. The parameter ε_e is not varied because, by Proposition 3, it does not affect any of the reported quantities. The first row (denoted by $\star$) displays the effects for the baseline combination of parameters. The remaining rows show the effect of varying each parameter separately, with the changed parameter highlighted in bold.

Table 6: Robustness Analysis

	Model parameters					Rates			Steady-state % changes		
	α	σ	β	A	ρ	τ^e	τ^w	s_K	ΔK	ΔH	ΔY
$\star$	0.33	0.80	0.90	10.00	−0.05	0.580	0.635	0.334	−16.87	+3.96	−3.49
2	**0.25**	0.80	0.90	10.00	−0.05	0.669	0.726	0.246	−21.15	+2.99	−3.53
3	**0.50**	0.80	0.90	10.00	−0.05	0.395	0.446	0.527	−13.05	+5.78	−4.58
4	0.33	**0.50**	0.90	10.00	−0.05	0.458	0.516	0.338	−11.81	+5.34	−0.79
5	0.33	**0.90**	0.90	10.00	−0.05	0.609	0.663	0.333	−21.28	−0.45	−7.90
6	0.33	0.80	**0.50**	10.00	−0.05	0.590	0.646	0.323	−17.27	+3.85	−3.48
7	0.33	0.80	**0.99**	10.00	−0.05	0.578	0.633	0.336	−16.81	+3.98	−3.50
8	0.33	0.80	0.90	**5.00**	−0.05	0.592	0.648	0.321	−17.34	+3.83	−3.48
9	0.33	0.80	0.90	**15.00**	−0.05	0.572	0.628	0.341	−16.61	+4.03	−3.51
10	0.33	0.80	0.90	10.00	**−0.01**	0.611	0.622	0.330	−2.92	+1.28	−0.13
11	0.33	0.80	0.90	10.00	**−0.10**	0.538	0.653	0.342	−38.12	+0.27	−14.76
12	0.33	0.80	0.90	10.00	**0.00**	0.619	0.619	0.330	0.00	0.00	0.00
13	0.33	0.80	0.90	10.00	**+0.05**	0.655	0.604	0.330	+10.83	−8.92	−2.85
14	0.33	0.80	0.90	10.00	**+0.10**	0.689	0.591	0.332	+15.39	−20.31	−10.01

$\star$ = baseline (Figures 6–8). In each row only the parameter in bold differs from the baseline; $\varepsilon_e = 0.20$ and $b_0 = 1$ throughout. τ^e and τ^w are the steady-state rates under the elite regime and under democracy; s_K is the income share of physical capital in the elite-regime steady state (it changes by less than 0.003 after the transition). ΔK, ΔH, ΔY: percentage change from the elite-regime to the democratic steady state. Rows 12–14 ($\rho \geq 0$) are reported for comparison with the baseline assumption $\rho < 0$. Source: own elaboration.

Several patterns emerge. First, for every parameterization with $\rho < 0$ (rows 1–11), the democratic transition raises the redistribution rate and lowers physical capital, while steady-state income falls. The change in human capital is positive in all but one case; in row 5, with $\sigma = 0.9$, the decline in bequests is strong enough to offset the higher share of bequests devoted to education, and long-run human capital falls slightly, even though it rises on impact. The income decline is modest when ρ is close to zero (row 10), because the two tax rates are nearly identical, and becomes substantial as the elasticity of substitution rises (row 11).

Second, the sign of ρ is decisive, as predicted by Proposition 2. With Cobb-Douglas technology (row 12), both groups choose the same tax rate and the democratic transition has no effect. With gross complements (rows 13–14), the elite chooses a *higher* rate than workers, so that democratization *lowers* redistribution, raises physical capital and lowers human capital, i.e., the opposite of the pattern observed in the data. Steady-state income falls in this case too, now because the workers' tax rate is too low relative to the wealth-maximizing rate. The assumption $\varepsilon_{K,H} > 1$ is therefore not a technicality, but

the condition that makes the model consistent with the evidence on the accumulation of physical and human capital following democratic transitions.

Third, the parameters α, β and A affect the magnitudes only moderately. A larger α lowers both rates and reduces the decline in physical capital while increasing the gain in human capital; β and A shift the level of the rates slightly but leave the percentage changes almost unchanged. The parameter σ matters more: a less concave education technology (higher σ) raises both rates, widens the decline in physical capital, and makes the long-run income loss larger.[11]

The income column should be read together with the factor shares. To a first-order approximation, $\Delta \ln Y \approx s_K \Delta \ln K + (1 - s_K) \Delta \ln H$, and the weight s_K differs across rows, so a smaller decline in K combined with a larger rise in H need not imply a smaller decline in Y. Comparing rows 2 and 3, for instance, physical capital falls less and human capital rises more when $\alpha = 0.50$ than when $\alpha = 0.25$. Yet, income falls more in the former case, since the capital share rises from 0.25 to 0.53 and the loss of physical capital receives more than twice the weight. The approximation reproduces the reported income changes to within 0.1 percentage points in every row. The same effect, on a much smaller scale, explains the ordering of rows 6–7 and 8–9: under gross substitutes, the capital share rises with the steady-state ratio K/H, which is higher when β or A are higher, so the decline in K in rows 7 and 9, although slightly smaller than in rows 6 and 8, is weighted slightly more heavily.

7 Conclusion

This paper presented an overlapping-generations model in which the economy's redistribution rate affects both physical capital accumulation (through a tax on bequests) and human capital accumulation (through public education financed by that tax), with a CES production function. The model was used to analyze the consequences of an exogenous democratic transition, understood as a shift from the elite-controlled redistribution rate to the rate chosen by the median voter, a worker.

The analysis delivers four main results. First, the direction of the political conflict over redistribution depends only on the elasticity of substitution between physical and human capital. Each group maximizes its own income, which is the income share of the factor it owns times total output; with gross substitutes, the share of the relatively abundant factor rises, so the elite prefers a lower rate than the output-maximizing one, while workers prefer a higher rate. Under Cobb-Douglas technology the two groups agree, and under gross complements the ordering reverses. The assumption of an elasticity of substitution

[11] In this case, the elite already chooses a high tax rate, given the high return of accumulating human capital. Workers then increase the tax rate even more, leading to further imbalance in the K/H ratio and to a relatively large income loss.

above one is therefore essential to reproduce the empirical pattern. Second, under gross substitutes, a democratic transition lowers physical capital and raises human capital, both on impact and, for most parameterizations, in the long run. This is consistent with the within-country evidence documented in Section 2 and with Tavares and Wacziarg (2001). Third, the share of capitalists in the population is neutral: it scales the economy without affecting redistribution rates or the percentage effects of the transition. Fourth, long-run income is lower under democracy in every parameterization considered. The elite, as the owner of the capital stock, chooses the rate that maximizes the after-tax return on bequests, which is also the rate that maximizes steady-state wealth; the median voter does not fully internalize the dynamic cost of a higher taxation.

The fourth result deserves emphasis because it qualifies the interpretation of the empirical literature. The model reproduces the negative effect of democracy on investment formalized by Persson and Tabellini (1994) and Alesina and Rodrik (1994), and the positive effect on human capital emphasized by Saint-Paul and Verdier (1993), but it shows that redirecting the proceeds of capital taxation to public education is not by itself sufficient to raise long-run income. The gains from democratization documented by Acemoglu et al. (2019), and the mildly positive income growth around transitions in our event study, must therefore operate through channels absent from the model. Natural candidates are reductions in rent extraction and in the deadweight costs of maintaining a non-democratic regime, lower social conflict, or productivity gains associated with a more educated population that are not captured by a constant technology level A.

Several extensions are left for future research. First, the democratic transition is treated as exogenous. Making it endogenous, for example by allowing capitalists and workers to bargain over the regime in the spirit of Acemoglu and Robinson (2006), would allow the model to generate predictions about when transitions occur and whether they are reversible. Second, introducing one of the channels mentioned above, such as a productivity externality from human capital or a resource cost of elite rule, would allow the model to be confronted with the evidence on income and would provide a measure of how large such a channel needs to be to overturn the accumulation effect. Third, extending the model to allow for a middle class that owns both physical and human capital, along the lines of Galor and Moav (2006), could generate richer predictions about the distributional consequences of democratization.

Our analysis also speaks to the specification of the aggregate production function. The empirical literature provides evidence against the Cobb–Douglas restriction (e.g., Duffy and Papageorgiou (2000)), but the evidence on the elasticity of substitution between physical and human capital remains unsettled. Some studies find evidence consistent with gross substitutability. For example, Duffy and Papageorgiou (2000) find that physical capital and human-capital-adjusted labor are more substitutable in their group of richest countries. Other studies, however, find evidence of complementarity between physical

capital and human capital or skilled labor. Krusell et al. (2000), for example, emphasize capital-skill complementarity, while Eisfeldt et al. (2023) provide estimates of the elasticity between physical and human capital that are below one. We therefore do not take a stand on a particular value of the elasticity of substitution. Instead, we emphasize the implications of the gross-substitutes case and show that it generates movements in physical and human capital that are consistent with patterns observed in the cross-country data. Although this evidence does not establish that physical and human capital are gross substitutes, it provides an additional motivation for considering this case and highlights its implications for the political economy of redistribution and human-capital accumulation.

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A Derivations and Proofs

Throughout, define the bracket

$$\Omega(\tau_t, b_t) \equiv \alpha\varepsilon_e^{-\rho}(1-\tau_t)^{-\rho}b_t^{-\rho} + (1-\alpha)(1-\varepsilon_e)^{-\rho(1-\sigma)}\varepsilon_e^{-\rho\sigma}\tau_t^{-\rho\sigma}b_t^{-\rho\sigma},$$

which is obtained by substituting (17) and (18) into $\alpha K_{t+1}^{-\rho} + (1-\alpha)H_{t+1}^{-\rho}$. Note that $W(\tau_t, b_t)$ in (24) equals $\frac{\rho+1}{\rho}\,\partial \ln\Omega/\partial\tau_t$.

A.1 Capitalist's Problem

The capitalist's problem for choosing the redistribution rate is:

$$\begin{aligned} \max_{\tau_t} \quad & (1-\beta)\ln c_{t+1}^e + \beta \ln b_{t+1} \\ \text{s.t.} \quad & c_{t+1}^e + b_{t+1} \le (1-\tau_t) r_{t+1} b_t \\ & b_{t+1} = \beta(1-\tau_t) r_{t+1} b_t \end{aligned}$$

Rewriting, one obtains:

$$\max_{\tau_t} (1-\beta)\ln\big[(1-\beta)(1-\tau_t)r_{t+1}b_t\big] + \beta\ln\big[\beta(1-\tau_t)r_{t+1}b_t\big]$$

Using equation (12) with $K_{t+1} = \varepsilon_e(1-\tau_t)b_t$,

$$\ln\big[(1-\tau_t)r_{t+1}\big] = \ln(A\alpha) + \ln(1-\tau_t) - (\rho+1)\ln\big[\varepsilon_e(1-\tau_t)b_t\big] - \frac{\rho+1}{\rho}\ln\Omega(\tau_t, b_t),$$

so the problem becomes

$$\max_{\tau_t}\; C_t - \rho\ln(1-\tau_t) - \frac{\rho+1}{\rho}\ln\Omega(\tau_t, b_t)$$

where

$$C_t \equiv (1-\beta)\ln(1-\beta) + \beta\ln\beta + \ln b_t + \ln(A\alpha) - (\rho+1)\ln(\varepsilon_e b_t).$$

The first-order condition is:

$$\frac{\rho}{1-\tau_t} - \frac{\rho+1}{\rho}\cdot\frac{\rho\alpha\varepsilon_e^{-\rho}(1-\tau_t)^{-\rho-1}b_t^{-\rho} - \rho\sigma(1-\alpha)(1-\varepsilon_e)^{-\rho(1-\sigma)}\varepsilon_e^{-\rho\sigma}\tau_t^{-\rho\sigma-1}b_t^{-\rho\sigma}}{\Omega(\tau_t, b_t)} = 0 \quad (29)$$

which is equation (23) in the main text.

A.2 Worker's Problem

The worker's problem for choosing the redistribution rate is:

$$\max_{\tau_t} \ \ln\big(w_{t+1}\, h_{t+1}\big)$$

Using equation (13) with $H_{t+1} = (1-\varepsilon_e)^{1-\sigma}\varepsilon_e^{\sigma}\tau_t^{\sigma}b_t^{\sigma}$ and $h_{t+1} = [\varepsilon_e\tau_t b_t/(1-\varepsilon_e)]^{\sigma}$,

$$\ln(w_{t+1}h_{t+1}) = \ln[A(1-\alpha)] - (\rho+1)\ln H_{t+1} - \frac{\rho+1}{\rho}\ln\Omega(\tau_t, b_t) + \sigma\ln\left[\frac{\varepsilon_e\tau_t b_t}{1-\varepsilon_e}\right].$$

The coefficient on $\ln\tau_t$ is $-(\rho+1)\sigma+\sigma = -\rho\sigma$, so the problem becomes

$$\max_{\tau_t} \ E_t - \rho\sigma\ln\tau_t - \frac{\rho+1}{\rho}\ln\Omega(\tau_t, b_t)$$

where

$$E_t \equiv \ln[A(1-\alpha)] - (\rho+1)\ln\big[(1-\varepsilon_e)^{1-\sigma}\varepsilon_e^{\sigma}b_t^{\sigma}\big] + \sigma\ln\big[\varepsilon_e(1-\varepsilon_e)^{-1}b_t\big].$$

The first-order condition is:

$$-\frac{\rho\sigma}{\tau_t} - \frac{\rho+1}{\rho}\cdot\frac{\rho\alpha\varepsilon_e^{-\rho}(1-\tau_t)^{-\rho-1}b_t^{-\rho} - \rho\sigma(1-\alpha)(1-\varepsilon_e)^{-\rho(1-\sigma)}\varepsilon_e^{-\rho\sigma}\tau_t^{-\rho\sigma-1}b_t^{-\rho\sigma}}{\Omega(\tau_t, b_t)} = 0 \quad (30)$$

which is equation (26) in the main text.

A.3 Proof of Proposition 2

Single-peakedness of output. From (11), $\ln Y_{t+1} = \ln A - \frac{1}{\rho}\ln\Omega(\tau_t, b_t)$. For $\rho < 0$, maximizing Y_{t+1} is equivalent to maximizing Ω, which is a sum of the two terms $c_1(1-\tau_t)^{|\rho|}$ and $c_2\tau_t^{\sigma|\rho|}$ with $c_1, c_2 > 0$; since $|\rho| \le 1$ and $\sigma|\rho| < 1$, both terms are strictly concave in τ_t, so Ω is strictly concave with $\Omega'(0^+) = +\infty$ and $\Omega'(1^-) = -\infty$, and has a unique interior maximizer τ_t^Y. For $\rho > 0$, maximizing Y_{t+1} is equivalent to minimizing Ω, which is now a sum of two strictly convex terms, $c_1(1-\tau_t)^{-\rho}$ and $c_2\tau_t^{-\rho\sigma}$, diverging at both endpoints, so it has a unique interior minimizer. For $\rho = 0$, $\ln Y_{t+1} = \text{const} + \alpha\ln(1-\tau_t) + (1-\alpha)\sigma\ln\tau_t$, which is strictly concave with maximizer $\tau^{\text{Cobb}} = \sigma(1-\alpha)/[\alpha + \sigma(1-\alpha)]$. In all cases Y_{t+1} is single-peaked in τ_t.

Shares. Since $Y_{t+1} = r_{t+1}K_{t+1} + w_{t+1}H_{t+1}$ (Euler's theorem), and using (17), (16) and (18),

$$(1-\tau_t)r_{t+1}b_t = \frac{r_{t+1}K_{t+1}}{\varepsilon_e} = \frac{s_K(\kappa_{t+1})Y_{t+1}}{\varepsilon_e}, \qquad w_{t+1}h_{t+1} = \frac{w_{t+1}H_{t+1}}{1-\varepsilon_e} = \frac{s_H(\kappa_{t+1})Y_{t+1}}{1-\varepsilon_e},$$

with s_K given in (14). Hence the elite's objective is $\ln s_K(\kappa_{t+1}) + \ln Y_{t+1}$ and the worker's objective is $\ln s_H(\kappa_{t+1}) + \ln Y_{t+1}$, up to constants.

Derivatives of the shares. From (14),

$$\frac{d\ln s_K}{d\ln\kappa} = -\rho + \frac{\rho\alpha\kappa^{-\rho}}{\alpha\kappa^{-\rho}+(1-\alpha)} = -\rho\,(1-s_K) = -\rho\,s_H, \qquad \frac{d\ln s_H}{d\ln\kappa} = -\frac{s_K}{s_H}\frac{d\ln s_K}{d\ln\kappa} = \rho\,s_K.$$

From (19), $d\ln\kappa_{t+1}/d\tau_t = -1/(1-\tau_t) - \sigma/\tau_t < 0$.

Ordering. At an interior solution τ_t^e of the elite's problem,

$$\frac{d\ln Y_{t+1}}{d\tau_t}\Big|_{\tau_t^e} = -\frac{d\ln s_K}{d\ln\kappa}\cdot\frac{d\ln\kappa_{t+1}}{d\tau_t} = \rho\,s_H\cdot\frac{d\ln\kappa_{t+1}}{d\tau_t},$$

which is strictly positive if $\rho < 0$, zero if $\rho = 0$, and strictly negative if $\rho > 0$. At an interior solution τ_t^w of the worker's problem,

$$\frac{d\ln Y_{t+1}}{d\tau_t}\Big|_{\tau_t^w} = -\rho\,s_K\cdot\frac{d\ln\kappa_{t+1}}{d\tau_t},$$

which is strictly negative if $\rho < 0$, zero if $\rho = 0$, and strictly positive if $\rho > 0$. Since Y_{t+1} is single-peaked with peak τ_t^Y, output is increasing to the left of τ_t^Y and decreasing to the right. For $\rho < 0$, output is increasing at τ_t^e and decreasing at τ_t^w, so $\tau_t^e < \tau_t^Y < \tau_t^w$. For $\rho > 0$ the inequalities reverse. For $\rho = 0$ both first-order conditions coincide with the output-maximization condition, so all three rates equal τ^{Cobb}. ■

A.4 Proof of Proposition 3

Define $\tilde{b}_t \equiv \varepsilon_e b_t/(1-\varepsilon_e)$. From (17) and (18),

$$K_{t+1} = (1-\varepsilon_e)(1-\tau_t)\tilde{b}_t, \qquad H_{t+1} = (1-\varepsilon_e)\tau_t^\sigma\tilde{b}_t^\sigma,$$

so $\kappa_{t+1} = (1-\tau_t)\tau_t^{-\sigma}\tilde{b}_t^{1-\sigma}$ does not depend on ε_e. Since r_{t+1} and w_{t+1} depend on (K_{t+1}, H_{t+1}) only through κ_{t+1}, and $h_{t+1} = \tau_t^\sigma\tilde{b}_t^\sigma$, both political objectives, $\ln[(1-\tau_t)r_{t+1}]$ and $\ln[w_{t+1}h_{t+1}]$, are functions of $(\tau_t, \tilde{b}_t)$ alone, and so are τ_t^e and τ_t^w. The bequest dynamics (20) imply $\tilde{b}_{t+1} = \beta(1-\tau_t)r(\kappa_{t+1})\tilde{b}_t$, again a function of $(\tau_t, \tilde{b}_t)$ alone. Hence the path of $\tilde{b}_t$, and the steady states $\tilde{b}^*$ under either regime, are independent of ε_e, while K, H and $Y = (1-\varepsilon_e)\,A[\alpha((1-\tau)\tilde{b})^{-\rho} + (1-\alpha)(\tau^\sigma\tilde{b}^\sigma)^{-\rho}]^{-1/\rho}$ are all proportional to $(1-\varepsilon_e)$. Ratios of steady-state values across regimes are therefore independent of ε_e. ■

A.5 Proof of Proposition 1

Fix τ and write $\kappa(b) = \Gamma(1-\tau)\tau^{-\sigma}b^{1-\sigma}$, which is strictly increasing in b, with $\kappa(0^+) = 0$ and $\kappa(\infty) = \infty$. Let $\phi(b) \equiv \beta(1-\tau)r(\kappa(b))$, so that $b_{t+1} = \phi(b_t)b_t$.

Monotonicity. From $\ln r(\kappa) = \ln(A\alpha) - (\rho+1)\ln\kappa - \frac{\rho+1}{\rho}\ln[\alpha\kappa^{-\rho} + (1-\alpha)]$,

$$\frac{d\ln r}{d\ln\kappa} = -(\rho+1) + (\rho+1)s_K(\kappa) = -(\rho+1)\,s_H(\kappa) \in \big(-(\rho+1),\,0\big),$$

so r is strictly decreasing in κ and ϕ is strictly decreasing in b. Moreover

$$\frac{d\ln[b\,\phi(b)]}{d\ln b} = 1 - (\rho+1)(1-\sigma)\,s_H(\kappa(b)) > 0,$$

since $(\rho+1)(1-\sigma) < 1$ for $\rho \le 0$ and $s_H < 1$; so $b \mapsto b\phi(b)$ is strictly increasing.

Limits. As $\kappa \to 0$ with $\rho < 0$, $\kappa^{-\rho} \to 0$ and $r(\kappa) \approx A\alpha(1-\alpha)^{-(\rho+1)/\rho}\kappa^{-(\rho+1)} \to \infty$. As $\kappa \to \infty$, $\kappa^{-\rho} \to \infty$ and $r(\kappa) \approx A\alpha\,\alpha^{-(\rho+1)/\rho}\kappa^{-(\rho+1)}\kappa^{\rho+1} = A\alpha^{-1/\rho}$. Hence $\phi(0^+) = \infty$ and $\phi(\infty) = \beta(1-\tau)A\alpha^{-1/\rho}$.

Existence, uniqueness and stability. Under condition (21), ϕ is continuous, strictly decreasing, and crosses 1 exactly once, at $b^* > 0$. For $b_t < b^*$ we have $\phi(b_t) > 1$, so $b_{t+1} > b_t$, and since $b\phi(b)$ is increasing, $b_{t+1} = b_t\phi(b_t) < b^*\phi(b^*) = b^*$; the sequence is increasing and bounded above by b^*, hence converges, and by continuity the limit is a fixed point, i.e. b^*. The case $b_t > b^*$ is symmetric. ■

A.6 Proof of Proposition 4

Let $G(\tau, b) \equiv \beta(1-\tau)\,r(\kappa(\tau,b))$ with $\kappa(\tau,b) = \Gamma(1-\tau)\tau^{-\sigma}b^{1-\sigma}$. By the proof of Proposition 1, $G_b < 0$. The steady state $b^*(\tau)$ is defined by $G(\tau, b^*(\tau)) = 1$, so by the implicit function theorem

$$\frac{db^*}{d\tau} = -\frac{G_\tau(\tau, b^*(\tau))}{G_b(\tau, b^*(\tau))},$$

whose sign is the sign of $G_\tau(\tau, b^*(\tau))$. Now the elite's objective at a given b is $\ln[(1-\tau)r_{t+1}] = \ln G(\tau,b) - \ln\beta$, so the elite's first-order condition (23) is exactly $G_\tau(\tau,b) = 0$. For $\rho < 0$ the derivative of the elite's objective, $\rho/(1-\tau) - W(\tau,b)$, tends to $+\infty$ as $\tau \to 0$ (the term in $\tau^{-\rho\sigma-1}$ in the numerator of W diverges while its denominator stays bounded, and $(\rho+1)/\rho < 0$) and to $-\infty$ as $\tau \to 1$. Since $\tau^e(b)$ is by assumption the unique interior critical point, we have $G_\tau(\tau,b) > 0$ for $\tau < \tau^e(b)$ and $G_\tau(\tau,b) < 0$ for $\tau > \tau^e(b)$. Hence $db^*/d\tau$ has the sign of $\tau^e(b^*(\tau)) - \tau$, which proves (i); at $\tau = \tau^{e*}$ this difference is zero.

For (ii), Proposition 2 gives $\tau^w(b) > \tau^e(b)$ for every b when $\rho < 0$. At the democratic steady state, $\tau^{w*} = \tau^w(b^*(\tau^{w*})) > \tau^e(b^*(\tau^{w*}))$, so by (i) $db^*/d\tau < 0$ at $\tau = \tau^{w*}$.

For (iii), define $\psi(\tau) \equiv \tau^e(b^*(\tau)) - \tau$. Under the stated condition $\psi'(\tau) < 0$ everywhere, so ψ is strictly decreasing and has at most one zero; since $\psi(\tau^{e*}) = 0$, we have $\psi > 0$ on $(0, \tau^{e*})$ and $\psi < 0$ on $(\tau^{e*}, 1)$. By (i), b^* is strictly increasing below τ^{e*} and strictly decreasing above it, which is the claim. Since $K^*(\tau) = \varepsilon_e(1-\tau)b^*(\tau)$ is the product of two positive decreasing functions on $(\tau^{e*}, 1)$, it is decreasing there as well. In the simulations of Sections 5 and 6, $\tau^e(b)$ changes by less than 0.01 per order of magnitude of b, so the condition in (iii) is comfortably satisfied. ■